LAB 8: Server Configuration – DHCP, DNS and Web Server in Cisco Packet Tracer

OBJECTIVES

- To configure a DHCP server to automatically assign IP addresses to client devices.
- To configure a DNS server to resolve domain names to IP addresses.
- To configure a Web Server to host web pages accessible by clients.
- To verify network functionality by testing connectivity and accessing web pages from client devices.

Software and Hardware Requirements

- Cisco Packet Tracer
- Windows Operating System
- Computer/Laptop
- Keyboard and Mouse

THEORY

In modern networks, servers play a critical role in providing essential services to clients. In this lab, we focus on three fundamental network services: DHCP, DNS, and Web Server.

DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addresses and other network settings, such as subnet mask, default gateway, and DNS server, to devices in a network. This eliminates the need to manually configure each device, which is inefficient and error-prone in large networks. In Packet Tracer, DHCP allows PCs and other devices to obtain IP addresses automatically from the server, simplifying network management.

DNS (Domain Name System)

DNS translates human-readable domain names, like www.example.com, into IP addresses that computers use to communicate. It makes network navigation easier because users do not have to remember numeric IP addresses. In Packet Tracer, a DNS server can resolve names for PCs within the network, enabling users to access services using domain names instead of IPs.

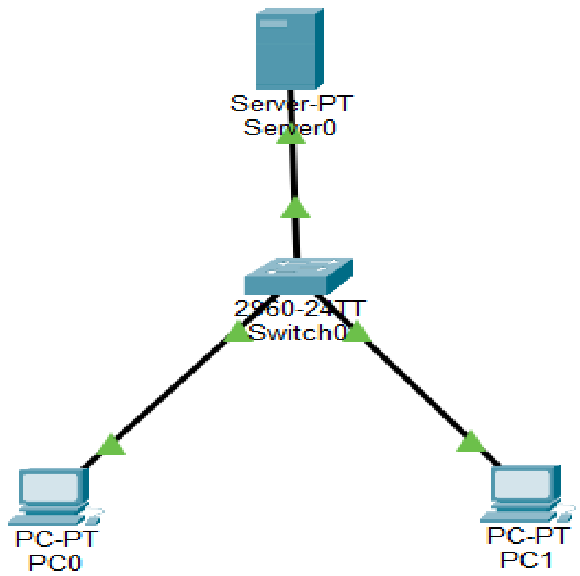
Web Server (HTTP/HTTPS)

A Web Server hosts website and serves web pages to clients upon request using protocols like HTTP or HTTPS. It allows users to access content via web browsers. In Packet Tracer, a Web Server can host simple HTML pages that client devices can access, providing a functional demonstration of web-based services in a network.

Integration of DHCP, DNS, and Web Server

When these services are combined, they make a network dynamic, user-friendly, and fully functional. A typical workflow in a network includes a PC booting up and receiving an IP address from the DHCP server, requesting a website via a domain name that the DNS server resolves to an IP, and finally retrieving the web page content from the Web Server. This demonstrates how these fundamental services work together to provide seamless network functionality.

NETWORK TOPOLOGY



CONFIGURATION

IP Addressing Scheme

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

For PC

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

For Server

Device	Default Gateway	DNS Server
Server0	192.168.1.1	192.168.1.2

PROCEDURE

Server Configuration

Step 1: Assign Static IP to Server

1. Click on Server
2. Go to Desktop → IP Configuration
3. Configure the following:
    - IP Address: 192.168.1.2
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
    - DNS Server: 192.168.1.2

DHCP Configuration

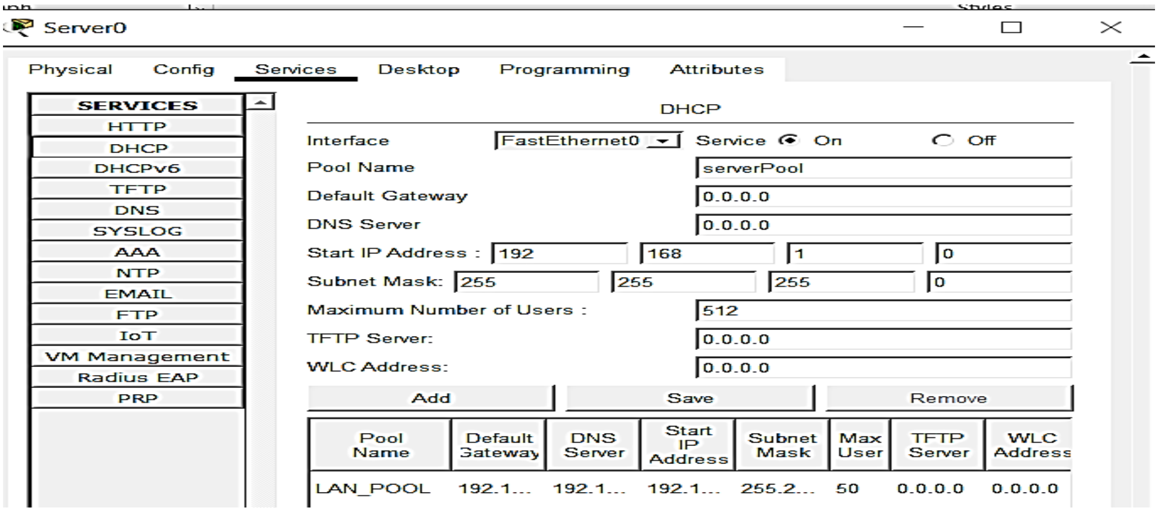
Step 2 : Enable DHCP Service

1. Go to Server → Services → DHCP
2. Turn DHCP: ON

Step 3 : Create DHCP Pool

Field	Value
Pool Name	LAB_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	0

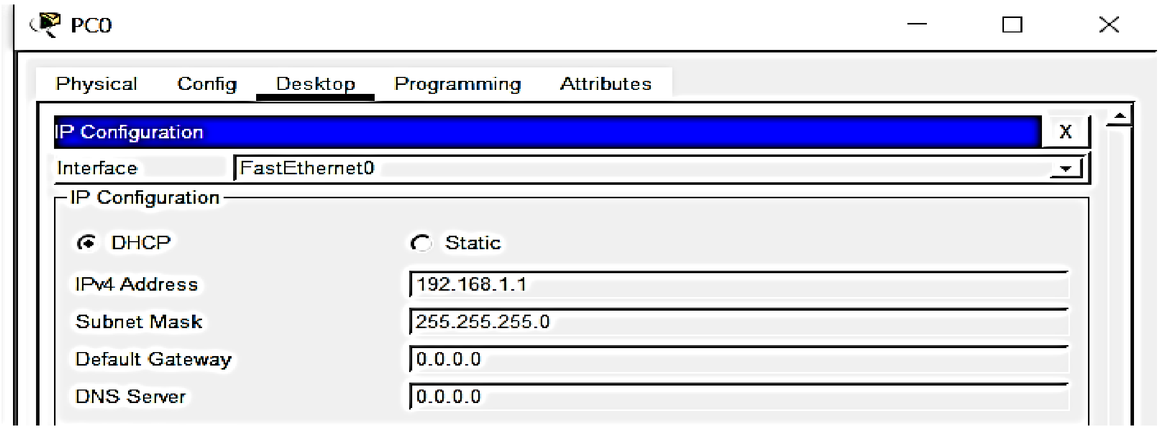
Click Add

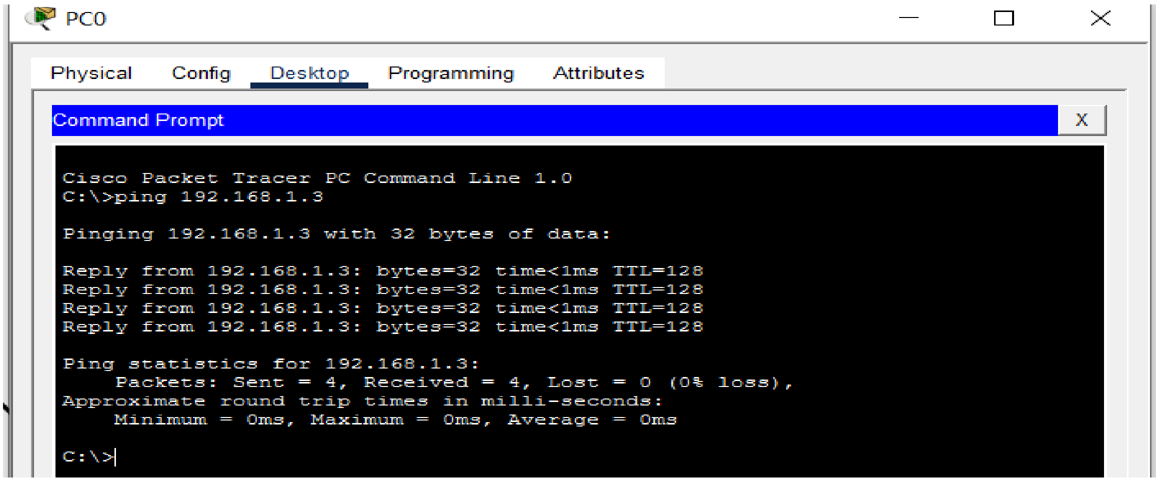


Verification

On each PC:

- Go to Desktop → IP Configuration
- Select DHCP





DNS Server Configuration

Step 4: Enable DNS Service

1. Go to Server → Services → DNS
2. Turn DNS: ON

Step 5: Add DNS Records

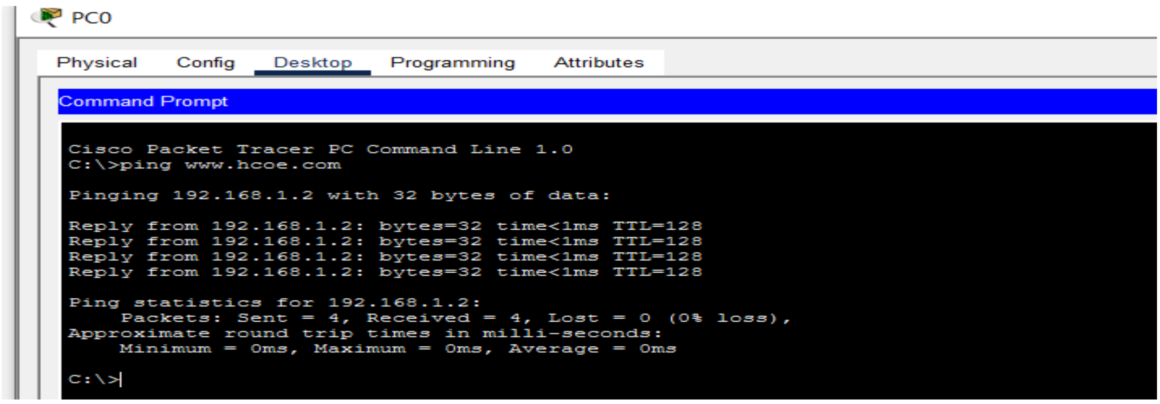
- Name: www.hcoe.com
- Type: A record
- Address: 192.168.1.2

Click Add

Verification

On each PC:

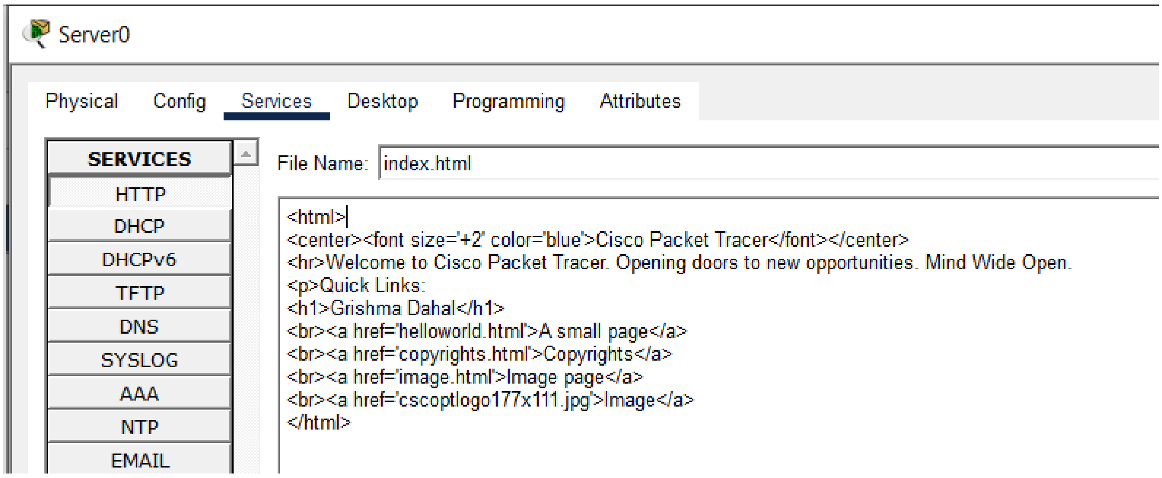
- Open Command Prompt
- Ping: www.hcoe.com



Web Server Configuration

Step 6: Enable HTTP Service

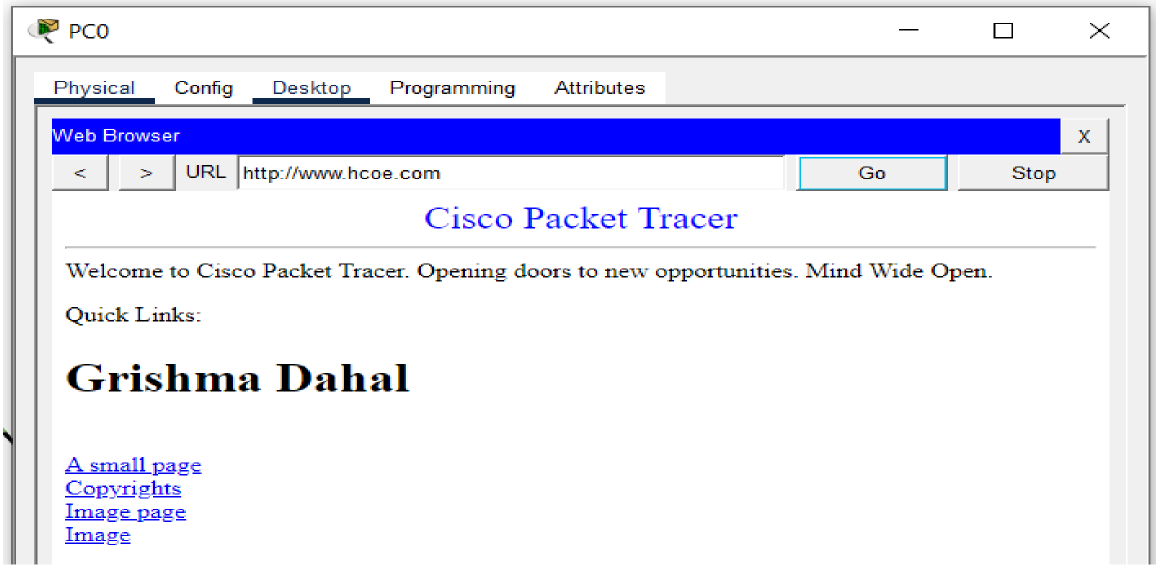
1. Go to Server → Services → HTTP
2. Turn HTTP: ON
3. Edit index.html (optional)



Verification

On PC:

- Go to Desktop → Browser
- Type: http://www.hcoe.com



RESULTS

The DHCP server was successfully configured with a pool named “LAB_POOL”, starting at IP 192.168.1.10, and client PCs set to DHCP mode automatically received valid IP addresses along with gateway information. A DNS “A record” was created for “www.hcoe.com” pointing to the server IP 192.168.1.2, and connectivity was verified by successfully pinging the domain name from client PCs. Additionally, the HTTP service was enabled on the server, allowing clients to access the hosted web content by entering “http://www.hcoe.com” into their web browsers.

DISCUSSION

This lab demonstrated the integration of DHCP, DNS, and Web Server services in a network. DHCP eliminated the need for manual IP configuration, reducing administrative work and the chance of human error. The DNS server provided a convenient way to access network resources by translating human-readable domain names into IP addresses. The workflow showed how these services interact: a PC first obtains an IP from DHCP, then uses DNS to resolve a URL, and finally connects to the Web Server to retrieve content. This synergy creates a dynamic and functional network environment, illustrating the importance of these services in modern networks.

CONCLUSION

The lab successfully showed the configuration and verification of essential server-side services in Cisco Packet Tracer. It confirmed that DHCP can automatically manage IP addresses, DNS can resolve domain names for easy navigation, and a Web Server can host content accessible to clients. Overall, testing verified that all services worked together seamlessly, providing a functional and user-friendly network experience.